

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458191.8665 +/- 0.004 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.169 +/- 0.049
Transit depth (Rp/Rs)^2	2.84 +/- 1.65 [%]
Orbital Inclination (inc)	84.19 +/- 1.82
Airmass coefficient 1 (a1)	0.966 +/- 0.099
Airmass coefficient 2 (a2)	0.026 +/- 0.031
Scatter in the residuals of the lightcurve fit is	10.11 %
Best Comparison Star	#2 - [155, 120]
Optimal Aperture	2.04
Optimal Annulus	14.48
Transit Duration (day)	0.042 +/- 0.045

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.169 ± 0.049 vs 0.1326 (deviation 0.74σ)
Duration fit vs published	0.042 d vs 0.1238 d (deviation 1.82σ)
Mid-time O–C	-6.7 min
Depth SNR	3.4
Residual scatter	10.11 %

Light curve

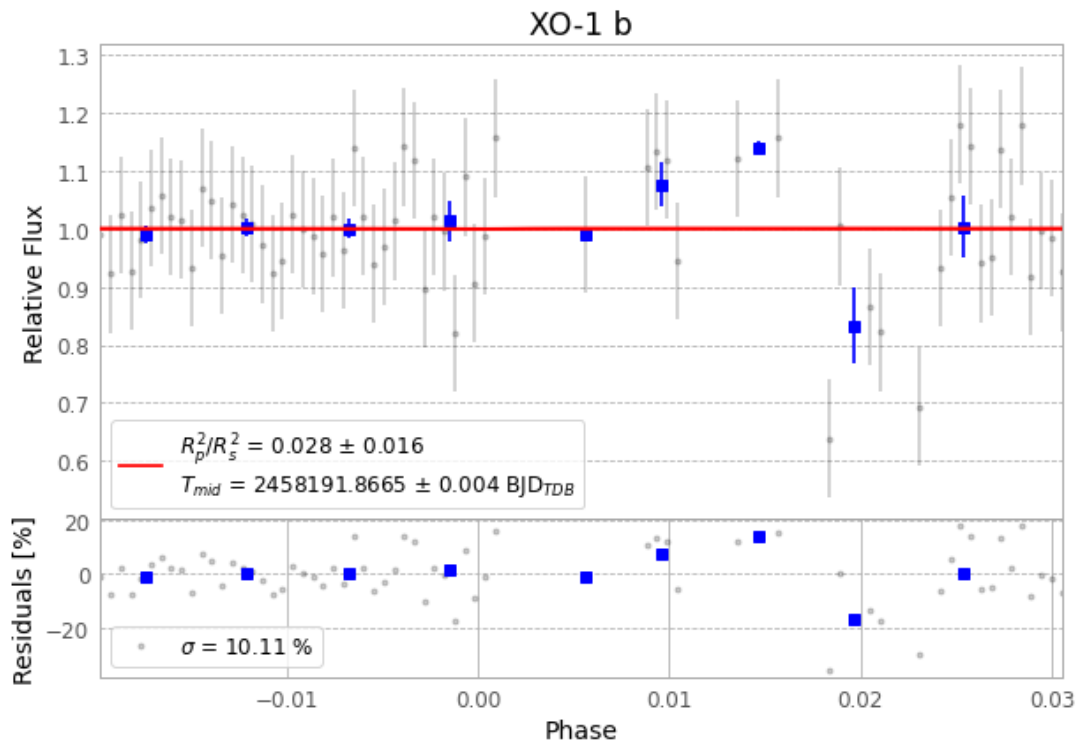


Figure 1 — FinalLightCurve_XO-1 b_2018-03-13.png (SHA-256 89b47c5ea41c3737...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [381, 169], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6fa4f6df2eed5422...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

94 FITS frames (62 MB), XO-1180314065112.FITS → XO-1180314113612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-XO-1-180314.log (f9b0f2f7253f...), FinalLightCurve_XO-1 b_2018-03-13.png (89b47c5ea41c...), FinalLightCurve_XO-1 b_2018-03-13.csv (da37ef4e555e...)

Compute resources

Wall time 380.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 03:54Z.